ADRIÀ TORRALBA-AGELL

Zero-Knowledge Cryptographer
Blockchain & Cryptography Security Researcher
PhD on Applied Cryptography on Blockchain

0xAdriaTorralba.me | crypto@0xAdriaTorralba.me | 0000-0002-1889-3404
0xAdriaTorralba | @0xAdriaTorralba | AdriaTorralba-Agell

INTERESTS

- Zero-Knowledge Cryptography
- Elliptic Curve Cryptography
- Homomorphic Encryption
- Abstract Algebra
- Blockchain Security
- Pairing-based Cryptography

TECH STACK

Solidity, Java, C++, C#, PIL2, Circom, Python, Javascript, Go, WebGL, Bash, R, OpenGL, LaTeX, PostgreSQL, Rust, MatLab, Assembler, Cairo, Docker

LANGUAGES

English: Advanced / B2 (FCE)

Spanish: Native / C1

Catalan: Native / C1

REFERENCES

Samuel Wong
Senior Blockchain Security Researcher
samuel-wong-asa
samwwong@alumni.stanford.edu

Felix Wegener
Security Research Manager
felix-wegener-7396115b
felix-wegener@gmx.net

Josep Chetrit
Lead Security Tooling Developer
josep-chetrit-1751621ab
josep.chetrit@openzeppelin.com

EXPERIENCE

Zero-Knowledge Cryptographer and Auditor | OpenZeppelin

Aug. 2025 - Feb. 2026 — Remote (Barcelona, Spain)

- Conducted security audit of ZisK zkVM, analysing cryptographic completeness and soundness.
- Developed expertise on Fully-Homomorphic Encryption.
- Researched and evaluated state-of-the-art zkVM architectures and their security trade-offs.
- Produced security and audit reports following industry best practices.

Blockchain ZK-Rollups Researcher | Migalabs

Oct. 2023 - May 2024 — Remote (Barcelona, Spain)

- Hands-on deploying real-world production layer 2 infrastructures.

PhD Candidate | Universitat Oberta de Catalunya

Dec. 2021 – Dec. 2025 — Barcelona, Spain

- Developed expertise on Zero-Knowledge Cryptography and Elliptic Curve Cryptography.
- Researched cryptographic primitives used on blockchain and blockchain security assumptions.
- Researched ZK-Rollup architectures, security properties and trust assumptions.

Teaching Assistant at Programming I & II | Universitat de Barcelona

Oct. 2019 – Jun. 2020 — Barcelona, Spain

- Taught Object-Oriented Programming (OOP) in Java.

EDUCATION

PhD in Applied Cryptography on Blockchain | Universitat Oberta de Catalunya

2021 – Present — Barcelona, Spain

MSc in Fundamental Principles of Data Science | Universitat de Barcelona

2020 – 2021 — Barcelona, Spain

Double BSc in Mathematics and Computer Science | Universitat de Barcelona

2013 – 2020 — Barcelona, Spain

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SELECTED PUBLICATIONS

Stefanos Chaliasos, Itamar Reif, Adrià Torralba-Agell, Jens Ernstberger, Assimakis Kattis, and Benjamin Livshits. "Analyzing and Benchmarking ZK-Rollups". In: *Advances in Financial Technologies* (2024)

Adrià Torralba-Agell and Cristina Pérez-Solà. "A Taxonomy of Security Analysis of Blockchain Layer 2 Scalability Solutions and Future Directions". In: *Blockchain: Research and Applications* (2024)

Adrià Torralba-Agell, Muoi Tran, and Cristina Pérez-Solà. "A Taxonomy and Security Analysis of Rollup Architectures in Ethereum Blockchain". In: *Reunión Española sobre Criptología y Seguridad de la Información* (2026)

Adrià Torralba-Agell, Ghazaleh Keshavarzkalhori, Cristina Pérez-Solà, David Megías, and Jordi Herrera-Joancomartí. "Unmasking the Illusion: The Shortcomings of "Zero-Knowledge" Rollups in Achieving Privacy". In: *Reunión Española sobre Criptología y Seguridad de la Información* (2024)

Adrià Torralba-Agell and Cristina Pérez-Solà. "A Comparison of Layer 2 Techniques for Scaling Blockchains". In: *Reunión Española sobre Criptología y Seguridad de la Información* 265 (2022)

SELECTED TALKS

Zero Knowledge for Privacy, Succinct for Scalability |

April 2024 — Privacy Reunion II

Hands-on Zero-Knowledge Basics - KZG Commitments |

Jul. 2023 — ETHBarcelona 2023

Security Analysis of Layer 1 and Layer 2 Extensions |

Feb. 2024 — Ethereum Dev Barcelona Meetup

A Comparison of Layer 2 Techniques for Scaling Blockchains |

Oct. 2022 — RECSI 2022, Santander (Spain)